

Panel Construction

§4.1

15 LLM Judges



▼ select K

Panel (K judges)

K_{eff} : MI-based diversity

Adversary Model

§3

Adaptive adversary

observes panel composition

6 Attack Types

Sycophancy

Authority

Verbosity

Score Manipulation

Position

Prompt Injection

**Target: exploit
weakest judge**

Vulnerability Metrics

§4.3

η_{max} **(weakest-link)**

max individual ASR

K_{eff} **(diversity)**

effective independent channels

Log-Linear Regression

$\log \text{ASR} \sim \log K_{\text{eff}} + \log \eta_{\text{max}}$

Aggregation

§4.4

Majority Vote

Weighted Vote

Threshold Vote

Bayesian Calib.

Robustness